

## Operating Systems

CSIT — 4th Semester — 2079 — Answer Sheet  
Subject Code: CSC264

**Note:** This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

### Section A

1. Discuss about single level and two-level directory system. Consider the following process and answer the following questions. 

|           |   |   |   |   |
|-----------|---|---|---|---|
| Process   | A | B | C | D |
| Max       | A | B | C | D |
| Available | A | B | C | D |
| P0        | 0 | 1 | 2 | 0 |
| P1        | 0 | 1 | 2 | 1 |
| P2        | 1 | 5 | 2 | 0 |
| P3        | 0 | 1 | 7 | 5 |
| P4        | 2 | 1 | 3 | 5 |
| P5        | 4 | 2 | 3 | 5 |
| P6        | 3 | 0 | 6 | 3 |
| P7        | 2 | 0 | 6 | 5 |
| P8        | 0 | 1 | 4 | 0 |
| P9        | 6 | 5 | 6 | A |

 A. What is the content of matrix Need? B. Is the system in safe state? C. If P1 request (0,4,2,0), can the request be granted immediately.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

2. When does race condition occur in inter process communication? What does busy waiting mean and how it can be handled using sleep and wakeup strategy?

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

3. Define shell and system call. Suppose a disk has 201 cylinders, numbered from 0 to 200. At same time the disk arm is at cylinder 95, and there is a queue of disk access requests for cylinders 82, 170, 43, 140, 24, 16 and 190. Calculate the seek time for the disk scheduling algorithm FCFS, SSTF, SCAN and C-SCAN.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

### Section B

Attempt any Eight questions [8×5=40]

4. Distinguish between starvation and deadlock. How does the system schedule process use multiple queues?

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

5. List any two demerits of disabling interrupt to achieve mutual exclusion. Describe about fixed and variable partitioning.

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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**6. For the following dataset, compute average waiting time for SRTN and SJF.ProcessArrival TimeBurst TimeP007P124P241P354**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**7. Discuss the advantages and disadvantages of implementing file system using Linked List.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**8. Consider the page references 7,0,1,2,0,3,0,4,2,3,0,3,2. Find the number of page fault using OPR and FIFO, with 4 page frame.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**9. Describe the working mechanism of DMA.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**10. What is the task of disk controller? List some drawback of segmentation.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**11. Write the structure and advantages of TLB.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

**12. Why do we need the concept of locality of reference? List the advantages and disadvantages of Round Robin algorithm.**

*Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]*

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*These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.*